FINAL BUSINESS INSIGHT REPORT

E-Commerce Sales Analysis

Data Analysis Project

ADITI SAHA

Tools Used

Microsoft Excel | MySQL | Power BI

Internship

[Primeor Solutions]

[19 October,2026]

02 | Project Overview

E-Commerce Sales Analysis

1. Project Objective

The objective of this project is to analyze e-commerce sales data to evaluate business performance and identify meaningful trends and patterns. The analysis focuses on sales, profitability, customer segments, product performance, and regional sales distribution. The project aims to identify top-performing products and customers, understand sales trends over time, and generate actionable business insights that can support data-driven decision-making.

2. Dataset Description

The project uses an e-commerce sales dataset containing order, customer, product, shipping, and financial information. The dataset includes approximately 51,271 records covering the period from 2011 to 2014.

Key attributes include Order ID, Order Date, Ship Date, Customer Name, Segment, Region, Market, Product Name, Category, Subcategory, Sales, Quantity, Discount, Profit, and Shipping Cost.

The dataset was examined and prepared for analysis by checking duplicate records, handling missing values, and reviewing date and numerical data types. These preparation steps helped improve data consistency and support reliable analysis.

3. Tools and Methodology

Microsoft Excel: Used for initial data inspection, cleaning, formatting, duplicate checks, and basic KPI calculations.

MySQL: Used to import and analyze the dataset through SQL queries, aggregations, filtering, and comparisons of sales and profitability across products, customers, regions, and categories.

Microsoft Power BI: Used to create interactive dashboards with KPI cards, sales trend charts, regional and market analysis, product and customer insights, and slicers for filtering the data.

Methodology: The project followed a structured workflow involving data preparation in Excel, data analysis using MySQL, and visualization through Power BI. The resulting findings were used to identify business trends and develop recommendations based on the analyzed data.

03 | EXCEL DATA ANALYSIS

E-Commerce Sales Analysis

1. Introduction

Microsoft Excel was used to inspect, clean, organize, and prepare the e-commerce dataset for further analysis.

2. Data Cleaning and Preparation

- **Duplicate Checking:** Checked duplicate records.
- **Missing Values:** Examined and handled blank values.
- **Data Formatting:** Reviewed date and numerical formats.
- **Data Validation:** Checked important sales and profit fields.

3. Key Performance Indicators (KPIs)

KPI	Value
Total Sales	12,625,222
Total Profit	1,467,497.62
Average Discount	14.29%
Total Rows	51,271

4. Analysis Summary

Excel was used to calculate initial KPIs and prepare the dataset for detailed analysis in MySQL and Power BI.

5. Outcome

The Excel stage provided an initial overview of sales performance and established a foundation for further business analysis.

04 | SQL Analysis

E-Commerce Sales Analysis

1. Introduction

MySQL was used to analyze e-commerce data and identify patterns in sales, profitability, customer spending, and regional performance.

2. SQL Analysis Performed

- **Sales Analysis:** Compared sales across regions, markets, and months.
- **Product Analysis:** Examined product sales and profitability by category and subcategory.
- **Customer Analysis:** Analyzed customer spending and identified high-value customers.
- **Profit Analysis:** Compared profits and identified loss-making orders.
- **Segment Analysis:** Evaluated sales across customer segments.

3. Key Business Insights

SQL queries helped examine regional sales performance, product profitability, customer contributions, and monthly sales trends. These findings can support the identification of business opportunities and areas requiring improvement.

4. Business Recommendations

- Focus on products with strong profitability.
- Investigate loss-making orders and review discounts and shipping costs.
- Improve regional sales performance.
- Use customer spending patterns to support customer retention.

5. Outcome

SQL analysis transformed raw data into meaningful summaries and provided a foundation for creating interactive Power BI dashboards and data-driven business recommendations.

05 | POWER BI Dashboard

E-Commerce Sales Analysis

1. Overview

An interactive E-Commerce Sales Dashboard was developed using Microsoft Power BI to analyze sales, profit, customer behaviour and product performance for 2011–2014.

2. Dashboard Pages

Executive Overview: Displays Total Sales, Total Profit, Total Orders and Average Discount through KPI cards.

- **Sales Analysis:** Visualizes sales by region, market, month and product category.
- **Product and Customer Insights:** Highlights top 10 products, top customers, profit by sub-category and segment analysis.

3. Interactive Features

Year, Region and Segment slicers allow users to filter data and explore different aspects of business performance. Consistent colours and chart formatting improve readability.

Conclusion

The Power BI dashboard presents the analysis through interactive visualizations, helping users understand sales trends, product performance, customer behaviour and profitability for informed business decision-making.

06 | BUSINESS INSIGHTS AND RECOMMENDATIONS

Project: E-Commerce Sales Analysis Using SQL, Excel and Power BI
Dataset Period: 2011–2014

1. Key Business Insights

Sales and Profit Performance
The analysis covers e-commerce sales transactions across multiple years, regions, markets, product categories and customer segments. The dashboard summarizes key performance indicators, including total sales, total profit, total orders and average discount, providing an overview of business performance.

Regional and Market Performance
The regional analysis identifies Central as the highest-sales region in the current dashboard, while APAC leads the displayed market comparison. These results help identify high-performing markets for further investigation.

Product and Category Performance
The product analysis highlights top-selling products and compares category-level sales. Evaluating sales alongside profit can help distinguish products that generate high revenue from those that contribute strongly to profitability.

Customer and Segment Analysis
Customer-level and segment-level analysis helps identify valuable customers and understand purchasing patterns across Consumer, Corporate and Home Office segments.

Discount and Profitability Analysis
Comparing discount levels with profit can help assess whether discounting supports sales growth or reduces profitability. Products and transactions with low or negative profit may require further investigation.

2. Business Recommendations

1. **Strengthen high-performing markets:** Investigate the factors behind strong regional and market sales and identify opportunities for sustainable growth.
2. **Optimize product performance:** Evaluate products using both sales and profit to support better inventory and product-promotion decisions.
3. **Improve discount strategies:** Review the relationship between discounts and profit before applying promotional offers.
4. **Focus on valuable customers:** Identify high-profit customers and customer segments to support retention and targeted marketing.
5. **Monitor profitability:** Regularly review low-profit and loss-making transactions to identify potential areas for improvement.

3. Conclusion

The project demonstrates how SQL, Excel and Power BI can be used together to analyze e-commerce data, monitor business performance and communicate findings through

interactive dashboards. The analysis supports data-driven decision-making by highlighting sales patterns, product performance, customer behaviour and profitability considerations.